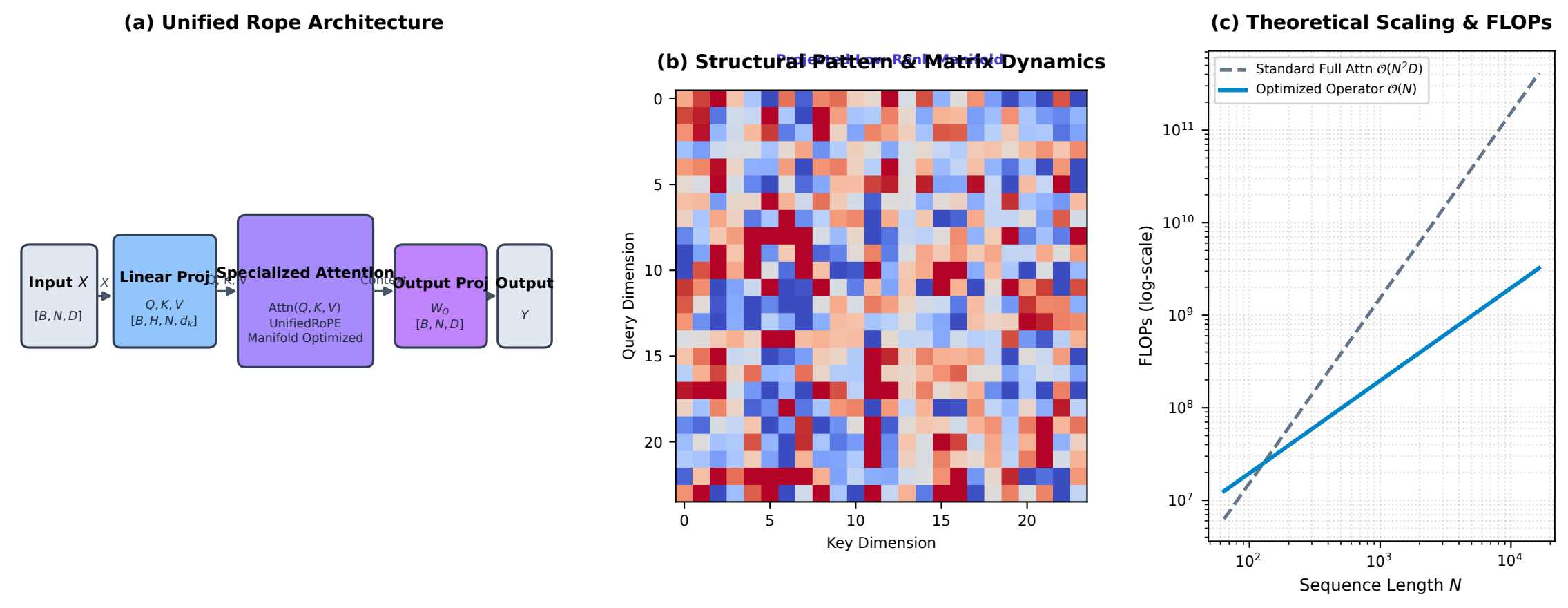
(a) Unified Rope Architecture
Input $X$
$[B, N, D]$
$X$
Linear Proj
$Q, K, V$
$[B, H, N, d_k]$
$Q, K, V$
Specialized Attention
Attn$(Q, K, V)$
UnifiedRoPE
Manifold Optimized
Context
Output Proj
$W_O$
$[B, N, D]$
Output
$Y$
(b) Structural Pattern & Matrix Dynamics
Query Dimension
Key Dimension
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Optimized Operator $\mathcal{O}(N)$
FLOPs (log-scale)
Sequence Length $N$
$10^7$
$10^8$
$10^9$
$10^{10}$
$10^{11}$
$10^2$
$10^3$
$10^4$